

TAPAS BARMAN

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linkedin

github

Portfolio



EXPERIENCE

MLOps Intern

Dec 2025 – Mar 2026

Aaizel International Technologies Private Limited

On-Site

- **(Climate Forte)** Built a production-grade **ETL & ML pipeline** using **Airflow** for data ingestion, model inference, and **S3** storage; automated retraining and drift monitoring via **MLflow, DVC, Evidently, and Jenkins**.
- **(VidyaDristi – OSINT)** Developed ML backend implementing **NER, LLM embeddings, vector search, and clustering** for intelligence extraction and entity resolution.

AI Software Research Intern

July 2025 – Dec 2025

Indian Institute of Technology Bombay (IIT Bombay)

Remote

- Worked on **Kalanjiyam**, a platform for **digital preservation and archiving** of Siddha literature through OCR-based text digitization and proofreading.
- **(Leap OCR)** Replaced **Tesseract** with **GPU-based EasyOCR**, achieving **3× speedup** and **18% error reduction**.
- **(Bhaasha OCR)** Built table detection and reconstruction using **YOLO + Transformers**, improving accuracy by **22%**.

Data Science Trainee

Sep 2024 – Jun 2025

Climate Resilience Observatory (CRO), Government of Uttar Pradesh

Lucknow, India

- **(Flood Risk Forecasting)** Built **XGBoost & Random Forest** models to predict flood inundation across 4 UP rivers, achieving **77% accuracy**; applied **SMOTE** for class imbalance and cleaned Hindi hydrological records to strengthen early warnings for **12 high-risk districts**.
- **(Lightning Alert System)** Engineered a **Python pipeline (JSON → SQL Server)** with **Folium & Plotly Dash** for real-time lightning visualization, enabling **8-minute alerts** via spatial-temporal analysis.

PROJECTS

Wind DSM Optimization

Python | XGBoost, Scikit-learn, MLflow | FastAPI, Apache Airflow, Docker | NOAA GFS, Pandas, NumPy

GitHub

- Built a **physics-informed ML pipeline** for a 75 MW wind plant, generating **96-block day-ahead forecasts** and improving accuracy by **71.2%**, reducing DSM penalties from **24.8L to 1.3L** and enabling projected annual savings of **1.41 Cr**
- Automated real-time forecasting using **Airflow and Docker**, ingesting NOAA GFS data and serving predictions via a **FastAPI REST API** with **MLflow-based model versioning** (Champion–Challenger)

AI Trend Notifier — Autonomous AI Trend Monitoring System

Python | LangGraph, Streamlit | Social APIs, HuggingFace, Tavily, Groq LLaMA, SQLite

GitHub

- Built an **autonomous agentic pipeline** using **LangGraph** to ingest AI trends from Twitter and Reddit, perform **DistilBERT-based sentiment analysis**, enrich content via **Tavily web search**, and generate summaries using a LLM.
- Developed a **production-grade web application** with a dedicated frontend and backend, enabling real-time trend visualization, subscriber management, and automated **email newsletter delivery** backed by **SQLite** and **SMTP**.

AI Medical Assistant

Python | LangGraph, LangChain, Flask | FAISS, HuggingFace, Groq LLaMA, Tavily, Europe PMC

GitHub

- Built a **multi-agent medical assistant** using a RAG pipeline capable of routing user queries, retrieving medical literature, and generating **evidence-based clinical responses**.
- Integrated **FAISS semantic search** with Europe PMC and Tavily APIs for real-time factual grounding, and deployed a **production-ready Flask backend** with persistent conversation history.

EDUCATION

Indian Institute of Information Technology, Lucknow

2024-2026

MSc Data Science - **SGPI - 8.85**

UP, Lucknow

TECHNICAL SKILLS

Languages & Systems: Python, SQL, Bash, Linux, Git, Docker, Docker Compose

MLOps & Data Engineering: MLflow (Tracking & Registry), DVC, Apache Airflow, Apache Kafka, Evidently AI

Generative AI: LangChain, LangGraph, RAG Pipelines, Tavily API, Groq LLaMA

Model Optimization: LoRA, QLoRA, Unsloth, Hugging Face Transformers, Quantization (bitsandbytes)

ML & Data Stack: PyTorch, Scikit-learn, Pandas, NumPy, FastAPI, MinIO/S3, SQLite

ACHIEVEMENTS

- **IIT-JAM 2024:** Secured All India Rank (AIR) 1200 in Physics
- **Active LeetCode Programmer:** Solved over **200+ problems**.
- **Sports Engagements:** Badminton, Chess, Cricket, Table Tennis